

No Show Paradox in Condorcet k -voting Procedures

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Abstract In this paper we extend the negative known results about No Show Paradox in Condorcet voting functions and correspondences to the contexts of k -functions and k -correspondences, in which the outcome of the voting process is a unique k -committee (set of k candidates) or a family of k -committees. The main result of the paper states that for every Condorcet k -function and for every Condorcet k -correspondence, there are situations in which every optimistic or pessimistic voter with some specific preferences could manipulate the election by abstaining.

Keywords Participation · Manipulation · Abstention paradox · Condorcet · k -functions voting rules · k -correspondences voting rules

1 Introduction

Many representative bodies in reality consist of a set of individuals: the committee of a company, a jury, the members of a court, and the nominees for some prize, among others. Whatever the case, it is clear that its constitution requires the selection of a specific number of candidates.

Therefore, a problem of increasing interest in the literature of Social Choice is the selection of a committee. More specifically it is interesting the study of the

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characteristics and properties of the different ways (usually voting methods) in which this selection takes place.

Some of these methods are those based on extensions of Condorcet Principle (if there is an alternative which beats all other alternatives in pairwise majority comparisons, then choose it) to k -functions.

In this sense, the most important extension of Condorcet Principle to the k -choice context (Gehrlein 1985) proposes to select, as the winner set, a set of k alternatives such that each element of this set defeats in pairwise majority comparisons to every alternative which does not belong to it, whenever such set exists (Condorcet Committee).

Based on the concept of weak k -Condorcet Committee *à la* Gehrlein, Ratliff (2003) discusses some inconsistencies that appear in some generalizations to k -functions of well-known Condorcet methods (like Dodgson and Kemeny methods) when selecting a k set of winners. On the other hand, Barberà and Coelho (2008) analyse the stability of some k -rules (based on Copeland, Dodgson, Maximin and Kemeny, among others) and establishes that only Kemeny k -rule always selects a weak Condorcet Committee whenever such set exists.

We focus our interest in this k -choice context analysing the existence of No Show Paradox or Abstention Paradox in k -functions and k -correspondences that select a Condorcet committee of fixed size k whenever such set exists.

More specifically, we would like to extend to the k -choice context the results for voting functions and correspondences establishing that every Condorcet method suffers from some kind of No Show Paradox, i.e. there are some voting situations in which some voters would achieve a better result by abstaining (see Moulin 1988; Pérez 2001; Jimeno et al. 2009).

We will use the Participation terminology. Participation properties exclude the possibility of certain situations in which a group of voter gains by abstaining. Thus, every time a specific version of Participation fails, a corresponding version of the Abstention Paradox appears.

The abstention paradox is a topic deserving close scrutiny and reflection. Many election processes held routinely in councils and parliaments around world use a variety of runoff voting methods that are affected by the paradox. On the other hand, a complete family of voting methods of high theoretical importance, the Condorcet methods, are also affected. Thus, as in the conventional 1-choice context (in which only one candidate will ultimately be elected), it is also natural to ask in this k -choice context what is the practical importance of the abstention paradox.

One may be tempted to think that the occurrence of the paradox is probably a rare event of little practical concern, but there are two reasons not to diminish its importance. The first reason, to be analysed in more detail in the last section, is that the occurrence does not seem to be quite rare an event. The second reason is that, no matter the frequency, the mere possibility of the paradox (a case in which the fact of voting is harmful for the voter) points to a deep flaw in the logic of the process of counting and aggregating votes. This flaw is worsened by the fact that, in general, the voters do not have the information needed to avoid incurring in the paradox.

The paper proceeds as follows. Section 2 presents the basic terminology, Sect. 3 states and proves the main results, Sect. 4 briefly discusses the practical relevance of these paradoxes and Sect. 5 makes some comments and conclusions.

2 Basic Terminology

Let $X = \{x_1, x_2, \dots, x_n\}$ be a set with two or more candidates. Preferences of any voter are supposed to be a complete order L over X . In general, we suppose this order to be linear (strict and complete). We say $L = x > y > z > t \dots$ to denote the preference linear order in which x is the most preferred candidate, y is the second one, and so on, and aLb means that a is preferred to b in L .

Given the set of candidates $X = \{x_1, x_2, \dots, x_n\}$, and any finite set $V = \{1, 2, \dots, m\}$ with one or more voters, we call a **Situation** any pair (X, p) , where p is a preference profile over X from V , that is to say, a m -tuple of linear orders over X , each one meaning the preferences of a voter over X .

Given any $k \in \{1, 2, \dots, \#X\}$, we denote by 2^X the power set of X and by $X_k \equiv \{A \in 2^X : \#A = k\}$ the family of subsets of X with cardinality k . From now on, suppose $k < n = \#X$.

Given a linear order of preferences L , we denote by E any order of preferences over the elements of the power set 2^X that are compatible with L , i.e. E represents any extension order of preference from L . For example, Leximax and Leximin are two possible extension orders that we can generate from L .

Let us call **k -voting function** (from now on **k -function**) any function f_k which maps any situation (X, p) to one, and only one, element of X_k . That is to say, $f_k(X, p) \in X_k$. $f_k(X, p)$ is the winner committee and the elements of $f_k(X, p)$ are the k chosen candidates (winners) constituting the elected committee from the preference profile p .

Similarly, let us call **k -voting correspondence** (from now on **k -correspondence**) any function f_k which maps any situation (X, p) to one, and only one, subset of X_k . That is to say, $f_k(X, p) \subset X_k$. $f_k(X, p)$ is now the family of potential winner committees from the preference profile p .

Given any X , any two disjoint sets of voters $V_1 = \{1, 2, \dots, m_1\}$ and $V_2 = \{m_1 + 1, m_1 + 2, \dots, m_1 + m_2\}$, and any two preference profiles p_1 and p_2 over X from, respectively, V_1 and V_2 , we can merge these two profiles in order to obtain a new profile over X from $V_1 \cup V_2$. This new profile will be called $p_1 + p_2$. Let us call $p - v$ to the profile obtained deleting a voter v from profile p .

Let (X, p) be any situation with n candidates and m voters. Given any two different candidates x, y from X , $p(x, y)$ is the number of voters which strictly prefer x to y in p . It is obvious that $p(x, y) + p(y, x) = m$. The square $n \times n$ matrix M_p , whose entries are $p(x, y)$ (and without entries in the main diagonal), will be called the **Comparison Matrix** for (X, p) .

Candidate x is said to **beat** y , denoted by $xW_p y$, if $p(x, y) > p(y, x)$, and x is said to **beat or tie** y if $p(x, y) \geq p(y, x)$. If, given a situation (X, p) , x beats any other candidate, then x is called the **Condorcet** candidate for this situation. A voting correspondence f is called **Condorcet** if it selects the Condorcet candidate as the only winner whenever exists.

Given a situation (X, p) and a set $A \subset X$, call DB_A the difference between the number of candidates not in A beaten by every member of A and the number of candidates not in A that beat every member of A , and call $\delta_A = \min \left\{ \min \{p(a, y)\}_{y \in X-A} \right\}_{a \in A}$.

In our intent to generalise the Condorcet criterion to k -functions we will consider the following concept in Gehrlein (1985): given a situation (X, p) , a set $A \in X_k$ is a **Condorcet k -Committee** or **Condorcet k -Set** if every candidate $x \in A$ beats any other candidate $y \in X - A$, i.e. $p(x, y) > p(y, x)$, $\forall x \in A \forall y \in X - A$.

A k -function f_k is called **Condorcet** if in every situation (X, p) with a Condorcet k -Committee A , this will be the winner set. In other words, $f_k(X, p) = A \in X_k$. In a similar way, a k -correspondence f_k is called **Condorcet** if, whenever a Condorcet k -Committee A exists, this will be the only winner set. In other words, $f_k(X, p) = \{A\} \subset X_k$.

Note that the Condorcet k -function and Condorcet k -correspondence are a natural extension of, respectively, Condorcet function and Condorcet correspondence concept (when $k = 1$ both concepts are identical).

We say that a k -function or k -correspondence f_k satisfies the **Weak Translation Invariance** property if $f_k(X, p) = f_k(X, p + q)$, for any two situations (X, p) and (X, q) where profile q consists on every possible linear order over X . It means that adding a set of voters with exactly balanced preferences to any other situation does not alter the voting result. It is a minimal property satisfied by almost every k -function and almost every k -correspondence.

3 Main Results

Following Jimeno et al. (2009) and García (2009) we define the concepts of participation to be used in this paper.

Definition 1 Given a k -function f_k and an order extension E , f_k satisfies **Participation in the sense of E** (or **E -participation**) if for every situation (X, p) , and for every group of voters v with identical preferences P , it is false that $f_k(X, p) E_P f_k(X, p + v)$.

In other words, in every situation (X, p) there is no group of identical voters v such that they prefer, in the sense of the extension E , the outcome $f_k(X, p)$ obtained when they abstain to the outcome $f_k(X, p + v)$ obtained when they vote (briefly, no voter prefers not to vote). We could say that **E -participation** excludes the abstention paradox for k -functions in the sense of E .

Definition 2 Given a k -correspondence f_k and an order extension concept E :

- (a) f_k satisfies **Moderate Participation in the sense of E** (or **E -moderate-participation**) if for every situation (X, p) , for every group of voters v with preferences P , and for every $A \in f_k(X, p)$, there exists $B \in f_k(X, p + v)$ such that it is false that $A E_P B$.
- (b) f_k satisfies **Weak Participation in the sense of E** (or **E -weak-participation**) if for every situation (X, p) , and for every group of voters v with preferences P , there exist $A \in f_k(X, p)$ and $B \in f_k(X, p + v)$ such that it is false that $A E_P B$.

In other words, in moderate case, for every situation (X, p) there is no voter v such that she prefers, in the sense of the extension E , any of the potential outcomes of $f_k(X, p)$ obtained when she abstains to every potential outcome of $f_k(X, p + v)$ obtained when she votes. In weak case, for every situation (X, p) there is no voter v such that she prefers, in the sense of the extension E , every potential outcome of $f_k(X, p)$ obtained when she abstains to every potential outcome of $f_k(X, p + v)$ obtained when she votes. We could say that E -moderate participation excludes a moderate abstention paradox for k -correspondences in the sense of E , while E -weak participation excludes a strong abstention paradox for k -correspondences in the sense of E .

Let us note that when a k -correspondence fails to satisfy Participation in the sense of E (moderate or weak) this means that there exist situations in which a group of identical voters could manipulate the election by abstaining.

Definition 3 Given a voter i , and a linear order $P : x_1 > x_2 > \dots > x_n$ expressing the preferences of this voter over the particular candidates in X , for every pair (A, B) of elements of X_k , $A = \{a_1, a_2, a_3, \dots, a_k\}$ and $B = \{b_1, b_2, b_3, \dots, b_k\}$, such that $A \neq B$, we define:

- (a) **OP (Ordering for Optimists):**
 $A \text{ OP}_P B$ if $\text{Max}_P A \text{ P Max}_P B$
 (The favourite candidate in A is strictly preferred to the favourite candidate in B)
- (b) **PE (Ordering for Pessimists):**
 $A \text{ PE}_P B$ if $\text{Min}_P A \text{ P Min}_P B$
 (The least preferred candidate in A is strictly preferred to the least preferred candidate in B)
- (c) **LexiMax (Leximax Ordering):**
 $A \text{ LexiMax}_P B$ if, being $a_1 \text{ Pa}_2 \text{ P} \dots \text{ Pa}_k$ and $b_1 \text{ Pb}_2 \text{ P} \dots \text{ Pb}_k$, there is an s such that $0 \leq s < k$, $a_1 = b_1, a_2 = b_2, \dots, a_s = b_s$ and $a_{s+1} \text{ Pb}_{s+1}$.
 (The most preferred candidate in A , a_1 , is strictly preferred to the most preferred candidate in B , b_1 ; or $a_1 = b_1$ and the second most preferred candidate in A , a_2 , is strictly preferred to the second most preferred candidate in B , b_2 ; ... and so on)
- (d) **LexiMin (Leximin Ordering):**
 $A \text{ LexiMin}_P B$ if, being $a_1 \text{ Pa}_2 \text{ P} \dots \text{ Pa}_h$ and $b_1 \text{ Pb}_2 \text{ P} \dots \text{ Pb}_k$, there is an s such that $0 < s \leq k$, $a_k = b_k, a_{k-1} = b_{k-1}, \dots, a_{k-s} = b_{k-s}$ and $a_{k-s+1} \text{ Pb}_{k-s+1}$.
 (The least preferred candidate in A , a_k , is strictly preferred to the least preferred candidate in B , b_k ; or $a_k = b_k$ and the second least preferred candidate in A , a_{k-1} , is strictly preferred to the second least preferred candidate in B , b_{k-1} ; ... and so on)
- (e) **SD (Strong Domination Ordering):**
 $A \text{ SD}_P B$ if for every $a \in A$ and $b \in B$, $a \text{ P } b$. (Every candidate in A is strictly preferred to every candidate in B)

Observe that OP , PE , $LexiMax$, $LexiMin$ and SD are asymmetric strict orders over X_k . We can also define the **OP and PE** order ($A \text{ OP and PE } B$ if $A \text{ OP } B$ and $A \text{ PE } B$), and the **LMM** order ($A \text{ LMM } B$ if $A \text{ LexiMax } B$ and $A \text{ LexiMin } B$).

In order to obtain the impossibility results we first need to establish an intermediate result.

Definition 4 Given a situation (X, p) , $x \in X$ is a lower-candidate in regard to a k -set $B \in X_k$, if $\alpha < \beta$, where $\alpha = \text{Max}\{p(x, b)\}_{b \in B}$ and $\beta = \text{Min}\{\text{Min}\{p(b, y)\}_{y \in X-B}\}_{b \in B}$. We will call x simply a lower-candidate if x is a lower-candidate in regard to some k -set.

Let us now propose the following lemma, which is an extension of known lemmas established in Moulin (1988) and Jimeno et al. (2009).

Lemma 1 Given any situation (X, p) and any Condorcet k -correspondence f_k , if x is a lower-candidate and:

- (a) f_k satisfies moderate **OP**-participation, or
 - (b) f_k satisfies moderate **PE**-participation and Weak Translation Invariance
- then for every $C \in f_k(X, p)$, $x \notin C$.

Proof

- (a) Suppose f_k is a Condorcet k -correspondence and let x be a lower-candidate in regard to the k -set $B = \{b_1, b_2, \dots, b_k\}$ in a situation (X, p) with m voters in the profile p . In this case, $\alpha < \beta$, where $\alpha = \text{Max}\{p(x, b)\}_{b \in B}$ and $\beta = \text{Min}\{\text{Min}\{p(b, y)\}_{y \in X-B}\}_{b \in B}$. In addition, let $\delta = \text{Min}\{p(b, x)\}_{b \in B}$. Note that if $\alpha = p(x, b_j)$ for some $b_j \in B$, then $\delta = p(b_j, x)$. Therefore $\alpha + \delta = m$. If we add a set of voters v with cardinality $(\delta - \beta)$ and preferences: $x > b_1 > b_2 > \dots > b_k > \dots$, to profile p we obtain a new profile $p' = p + v$ with $m' = m + \delta - \beta$ voters. We will prove that, in the new situation (X, p') , B is a Condorcet k -set.

In the new situation and for all $b \in B$, we have $p'(b, x) = p(b, x)$ and $p'(x, b) = p(x, b) + \delta - \beta$. We know that $\delta < p(b, x)$ and $\beta > p(x, b)$, therefore $p'(x, b) = p(x, b) + \delta - \beta < p(b, x) = p'(b, x)$. Then $p'(b, x) > m'/2$. Additionally, for all $b \in B$ and for all $y \in X - B - \{x\}$, $p'(b, y) = p(b, y) + \delta - \beta$. Thus the expression $p'(b, y) > m'/2$ is equivalent to the expression $p(b, y) + \delta - \beta > (m + \delta - \beta)/2$, i.e. $2p(b, y) + \delta - \beta > m$. And this is equivalent to expression $p(b, y) + \delta + p(b, y) - \beta > m$ that we know is correct because $p(b, y) \geq \beta > \alpha$ and $\alpha + \delta = m$. Therefore $p'(b, y) > m'/2, \forall b \in B$.

Hence, in the new profile p' , B is a Condorcet k -set and then the lower-candidate x doesn't belong to the k -winner set.

In conclusion, because $f_k(X, p') = \{B\}$ and $\text{Sup}_v B <_v x$, if f_k satisfies **OP**-participation then for every $C \in f_k(X, p)$, $x \notin C$.

- (b) Suppose f_k is a Condorcet k -correspondence and let x a lower-candidate in regard to the k -set $B = \{b_1, b_2, \dots, b_k\}$ in a situation (X, p) with m voters and n candidates ($n > k$). Now, as in case (a), we have $\alpha < \beta$, where $\alpha = \text{Max}\{p(x, b)\}_{b \in B}$ and $\beta = \text{Min}\{\text{Min}\{p(b, y)\}_{y \in X-B}\}_{b \in B}$. Let $\delta = \text{Min}\{p(b, x)\}_{b \in B}$ and suppose that $(\delta - \beta)$ times a set of $n!$ new voters is added to the situation, whose preferences are all permutation orders of the n

candidates. In the new profile p' , with $m + (\delta - \beta)n!$ voters, $p'(z, t) = p(z, t) + (\delta - \beta)n!/2$, $\forall z, t \in X$, and thus $\alpha' = \alpha + (\delta - \beta)n!/2$, $\beta' = \beta + (\delta - \beta)n!/2$ and $\delta' = \delta + (\delta - \beta)n!/2$. Therefore, $\alpha' = \text{Max}\{p'(x, b)\}_{b \in B} < \beta' = \text{Min}\{\text{Min}\{p'(b, y)\}_{y \in X-B}\}_{b \in B}$. If now we delete $(\delta - \beta)$ identical voters with preferences: $b_1 > b_2 > \dots > b_k > x$, the result is a new profile p'' with $m'' = m' - (\delta - \beta)$ voters. We will prove that, as in case (a), in this new situation (X, p'') , B is a Condorcet k -set.

For all $b \in B$, $p''(b, x) = p'(b, x) - (\delta' - \beta') = p'(b, x) - (\delta - \beta)$ and $p''(x, b) = p'(x, b)$. And giving that $\delta' < p'(b, x)$ and $\beta' > p'(x, b)$, we deduce that $p''(x, b) = p'(x, b) < p'(b, x) - \delta' + \beta' = p'(b, x) - (\delta' - \beta') = p''(b, x)$. Therefore $p''(b, x) > m''/2$.

In addition, for all $b \in B$ and for all $y \in X - B - \{x\}$, $p''(b, y) = p'(b, y)$ and $p''(y, b) = p'(y, b) - (\delta' - \beta')$. Thus, the following expressions are equivalents:

- (1) $p''(b, y) > m''/2$,
- (2) $p'(b, y) > p'(y, b) - (\delta' - \beta')$,
- (3) $\delta' - \beta' + 2p'(b, y) > m'$ and
- (4) $p'(b, y) + \delta' + p'(b, y) - \beta' > m$

All of them are correct because in the last one we know that $p'(b, y) \geq \beta' > \alpha'$ and $\alpha' + \delta' = m'$. Therefore, $p''(b, y) > m''/2$, $\forall y \in X - B - \{x\}$.

Hence, in the new profile p'' , B is a Condorcet k -set and this implies that the lower-candidate x is not included in the k -winner set $f_k(X, p'')$ by any Condorcet k -correspondence.

This means that in the profile p' , where the $(\delta - \beta)$ identical voters with preferences: $b_1 > b_2 > \dots > b_k > x$ are not deleted, x doesn't belong to the k -winner set, $f_k(X, p')$, because in other case, by Weak Translation Invariance property, this implies that in the profile p , $x \in f_k(X, p)$ which is a failure of **PE-participation** property. Therefore, for every $C \in f_k(X, p)$, $x \notin C$. $\square$

Thus, there is no lower-candidate that belongs to any of the potential k -winner sets of a Condorcet k -correspondence.

Proposition 1 For every $k \in \{1, 2, 3, \dots\}$:

- (a) All Condorcet k -correspondences fail to satisfy moderate **OP-participation**.
- (b) All Condorcet k -correspondences satisfying Weak Translation Invariance fail to satisfy moderate **PE-participation**.

Proof Let $X = \{x_1, x_2, x_3, x_4, x_5\}$, with $m = 15$ voters, a set X with $n = 5k$ candidates $\{a_1 \dots a_k, b_1 \dots b_k, c_1 \dots c_k, d_1 \dots d_k, e_1 \dots e_k\}$ and a preference profile p : $[[b_1 \dots b_k] > [a_1 \dots a_k] > [c_1 \dots c_k] > [e_1 \dots e_k] > [d_1 \dots d_k]$ (4 voters), $[d_1 \dots d_k] > [a_1 \dots a_k] > [b_1 \dots b_k] > [c_1 \dots c_k] > [e_1 \dots e_k]$ (5 voters), $[e_1 \dots e_k] > [d_1 \dots d_k] > [b_1 \dots b_k] > [c_1 \dots c_k] > [a_1 \dots a_k]$ (3 voters), $[e_1 \dots e_k] > [d_1 \dots d_k] > [c_1 \dots c_k] > [b_1 \dots b_k] > [a_1 \dots a_k]$ (3 voters)], where the preference orders between elements of each k -set $A = \{a_1, \dots, a_k\}$, $B = \{b_1, \dots, b_k\}$, $C = \{c_1, \dots, c_k\}$, $D = \{d_1, \dots, d_k\}$ and $E = \{e_1, \dots, e_k\}$ are arbitrary.

The inter-element comparison matrix of k -set is M_p :

	$a_1 \dots a_k$	$b_1 \dots b_k$	$c_1 \dots c_k$	$d_1 \dots d_k$	$e_1 \dots e_k$
$a_1 \dots a_k$		5	9	4	9
$b_1 \dots b_k$	10		12	4	9
$c_1 \dots c_k$	6	3		4	9
$d_1 \dots d_k$	11	11	11		5
$e_1 \dots e_k$	6	6	6	10	

Suppose f_k is a Condorcet k -correspondence that satisfies **OP**-A-participation property, or satisfies **PE**-A-participation and Weak Translation Invariance properties. Observe that from Lemma 1:

$$\begin{aligned} \forall b_i \in B : \text{Max}\{p(b_i, d)\}_{d \in D} = 4 < 5 = \text{Min}\{\text{Min}\{p(d, x)\}_{x \in X-D}\}_{d \in D} &\Rightarrow b_i \notin f_k(X, p) \\ \forall c_i \in C : \text{Max}\{p(c_i, d)\}_{d \in D} = 3 < 4 = \text{Min}\{\text{Min}\{p(b, x)\}_{x \in X-B}\}_{d \in D} &\Rightarrow c_i \notin f_k(X, p) \\ \forall a_i \in A : \text{Max}\{p(a_i, d)\}_{d \in D} = 4 < 5 = \text{Min}\{\text{Min}\{p(d, x)\}_{x \in X-D}\}_{d \in D} &\Rightarrow a_i \notin f_k(X, p) \\ \forall d_i \in D : \text{Max}\{p(d_i, d)\}_{d \in D} = 5 < 6 = \text{Min}\{\text{Min}\{p(d, x)\}_{x \in X-E}\}_{d \in D} &\Rightarrow d_i \notin f_k(X, p) \end{aligned}$$

Therefore, $f_k(X, p) = \{E\}$.

If we add a new profile v of 4 identical voters with preference order $\{c_1 \dots c_k\} > \{e_1 \dots e_k\} > \{a_1 \dots a_k\} > \{b_1 \dots b_k\} > \{d_1 \dots d_k\}$, the new inter-element comparison matrix is M_{p+v} :

	$a_1 \dots a_k$	$b_1 \dots b_k$	$c_1 \dots c_k$	$d_1 \dots d_k$	$e_1 \dots e_k$
$a_1 \dots a_k$		9	9	8	9
$b_1 \dots b_k$	10		12	8	9
$c_1 \dots c_k$	10	7		8	13
$d_1 \dots d_k$	11	11	11		5
$e_1 \dots e_k$	10	10	6	14	

From Lemma 1 applied to situation $(X, p + v)$, candidates belonging to sets C , D and E are not chosen, because they are lower-candidates (all $c_i \in C$ are lower-candidates in regard to B , all $d_i \in D$ are lower-candidates in regard to E and all $e_i \in E$ are lower-candidates in regard to C). Thus every k -winner set $C \in f_k(X, p')$ in profile p' satisfies $C \subset A \cup B$. But this result contradicts the hypothesis that f_k satisfies k -participation in the sense of **OP**, because when these 4 voters are added one after another, at some point the new voter will see that she prefers in the sense of **OP** the outcome E obtained had she abstained to any of the k -sets, contained in $A \cup B$, obtained through voting. We can argue in a similar way that it also contradicts the hypothesis that f_k satisfies k -participation in the sense of **PE** and Weak Translation Invariance. $\square$

Proposition 2 For every $k \in \{1, 2, 3, \dots\}$:

- (a) All Condorcet k -functions fail to satisfy **OP**-participation
- (b) All Condorcet k -functions satisfying Weak Translation Invariance fail to satisfy **PE**-participation.

Proof It is a direct corollary of Proposition 1, because k -functions are just k -correspondences with a unique k -winner set. $\square$

Participation in the sense of E implies participation in the sense of every order E' stronger than (or included in) E , because if some pair (A, B) , where $A \in f(X, p)$ and $B \in f(X, p + v)$, is excluded of the relation E by definition of E -participation, it will also be excluded of the relation E' . Therefore, if E' implies E , then failing of E' -participation implies failing of E -participation. As a consequence, part (a) of Propositions 1 and 2 remains valid if the **OP** order is replaced by any extension order implied by **OP**, like the order **LexiMax**. In a similar way, part (b) of Propositions 1 and 2 remains valid if the **PE** order is replaced by any extension order implied by **PE**, like the order **LexiMin**.

Example 1 Let us see two particular examples of k -correspondences failing to satisfy some Participation properties. The example will be worded for $k = 2$, but could be easily replicated for every specific $k \in \{1, 2, 3, \dots\}$. Let us define the k -correspondences f_k^{COPELAND} and f_k^{MINMAX} , which are natural extensions to the k -choice context of the well-known Condorcet correspondences COPELAND and MINMAX.

Given any situation (X, p) with n candidates, and any $k \in \{1, 2, 3, \dots\}$ such that $k < n$.

$$f_k^{\text{COPELAND}}(X, p) \equiv \{A \in X_k \mid \text{DB}_A \geq \text{DB}_B \text{ for every } B \in X_k\}$$

$$f_k^{\text{MINMAX}}(X, p) \equiv \{A \in X_k \mid \delta_A \geq \delta_B \text{ for every } B \in X_k\}$$

It easy to see that f_k^{COPELAND} and f_k^{MINMAX} are Condorcet k -correspondences.¹ Let now (X, p) be the situation used in Proposition 1 and $k = 2$. In order to have a well defined detailed comparison matrix, we also suppose that every voter in the example prefers a_1 to a_2 , b_1 to b_2 , c_1 to c_2 , d_1 to d_2 and e_1 to e_2 . The matrix is:

	a_1	a_2	b_1	b_2	c_1	c_2	d_1	d_2	e_1	e_2
a_1		15	5	5	9	9	4	4	9	9
a_2	0		5	5	9	9	4	4	9	9
b_1	10	10		15	12	12	4	4	9	9
b_2	10	10	0		12	12	4	4	9	9
c_1	6	6	3	3		15	4	4	9	9
c_2	6	6	3	3	0		4	4	9	9
d_1	11	11	11	11	11	11		15	5	5
d_2	11	11	11	11	11	11	0		5	5
e_1	6	6	6	6	6	6	10	10		15
e_2	6	6	6	6	6	6	10	10	0	

$B = \{b_1, b_2\}$ and $D = \{d_1, d_2\}$, with $\text{DB}_B = \text{DB}_D = 4$, are the only 2-sets whose DB counts are maximal. Thus, $f_2^{\text{COPELAND}}(X, p) \equiv \{B, D\}$. If we add a new profile v of 4 identical voters with preference order $d_1 > d_2 > e_1 > e_2 > a_1 > a_2 > b_1 > b_2 > c_1 > c_2$, the new comparison matrix is:

¹ These extensions are not the only ones. Barberà and Coelho (2008) shows other extensions of Copeland and Minmax procedures to k -choice context. However, in the case of Minmax procedures, the showed methods are not Condorcet k -correspondences.

	a_1	a_2	b_1	b_2	c_1	c_2	d_1	d_2	e_1	e_2
a_1		19	9	9	13	13	4	4	9	9
a_2	0		9	9	13	13	4	4	9	9
b_1	10	10		19	16	16	4	4	9	9
b_2	10	10	0		16	16	4	4	9	9
c_1	6	6	3	3		19	4	4	9	9
c_2	6	6	3	3	0		4	4	9	9
d_1	15	15	15	15	15	15		19	9	9
d_2	15	15	15	15	15	15	0		9	9
e_1	10	10	10	10	10	10	10	10		19
e_2	10	10	10	10	10	10	10	10	0	

Since $E = \{e_1, e_2\}$ is now the Condorcet 2-committee ($DB_E = 8$), it is the only chosen 2-committee. Thus, $f_2^{\text{COPELAND}}(X, p + v) \equiv \{E\}$. Observe that, because the element $D \in f_2^{\text{COPELAND}}(X, p)$ satisfies $DOP_v S$ and $DPE_v S$ for every $S \in f_2^{\text{COPELAND}}(X, p + v)$, moderate OP -participation and moderate PE -participation fail. In fact, this example shows that f_k^{COPELAND} falls into a stronger abstention paradox. Indeed, f_k^{COPELAND} also fails to satisfy moderate participation in the sense of the order SD , a stronger order relation than OP and than PE . Furthermore, f_k^{COPELAND} not only fails to exclude the existence of an element $D \in f_k^{\text{COPELAND}}(X, p)$ such that $DOP_v S$ for every $S \in f_k^{\text{COPELAND}}(X, p + v)$, but even fails to exclude the case in which such an element D is constituted by the k most preferred candidates of v .

Let us analyse now f_k^{MINMAX} for the same situation (X, p) . It is easy to see that $f_2^{\text{MINMAX}}(X, p) = \{E\}$. If we add a new profile v of 4 identical voters with preference order $c_1 > c_2 > e_1 > e_2 > a_1 > a_2 > b_1 > b_2 > d_1 > d_2$, we obtain the following new comparison matrix M_{p+v} shown (abbreviated) above.

	a_1	a_2	b_1	b_2	c_1	c_2	d_1	d_2	e_1	e_2
a_1		19	9	9	9	9	8	8	9	9
a_2	0		9	9	9	9	8	8	9	9
b_1	10	10		19	12	12	8	8	9	9
b_2	10	10	0		12	12	8	8	9	9
c_1	10	10	7	7		19	8	8	13	13
c_2	10	10	7	7	0		8	8	13	13
d_1	11	11	11	11	11	11		19	5	5
d_2	11	11	11	11	11	11	0		5	5
e_1	10	10	10	10	6	6	14	14		19
e_2	10	10	10	10	6	6	14	14	0	

Let $H = \{a_1, b_1\}$. Since $\delta_A = \delta_B = \delta_H = 8$, it is clear that A, B and H are the only 2-winner committees in $(X, p + v)$. Thus, $f_2^{\text{MINMAX}}(X, p) = \{E\}$ and $f_2^{\text{MINMAX}}(X, p + v) = \{A, B, H\}$. It is clear that there is no $S \in f_2^{\text{MINMAX}}(X, p)$ and no $T \in f_2^{\text{MINMAX}}(X, p + v)$ such that it is false $SOP_v T$ or it is false $SPE_v T$. Therefore, f_k^{MINMAX} fails to satisfy even weak OP -participation (and also weak PE -participation). In fact, f_k^{MINMAX} also fails to satisfy weak participation in the sense of the order SD .

Propositions 1 and 2 establish an impossibility result for all Condorcet k -correspondences and k -functions. We could ask if this result is the strongest that can be achieved. More specifically: Can we strengthen Proposition 1 replacing moderate

OP-participation by weak **OP**-participation or by moderate **OP** and **PE**-participation? Can we obtain an analog result to Proposition 1 replacing **OP**-participation by **LMM**-participation? As we will prove in Proposition 3, using the Condorcet k -correspondence f_k^1 , the answer to both questions is negative.

Let us define f_k^1 . Given any situation (X, p) ,

$$f_k^1(X, p) \equiv \begin{cases} \{C\}, & \text{if there exists a Condorcet } k - \text{committee } C \\ X_k, & \text{if there is no Condorcet } k - \text{committee} \end{cases}$$

Proposition 3 *For every $k \in \{1, 2, 3, \dots\}$, the Condorcet k -correspondence f_k^1 satisfies weak LexiMax-participation and weak LexiMin-participation.*

Proof Given any situation (X, p) and any set v of identical voters, we will show that there is no case in which for every $S \in f_k^1(X, p)$ and $T \in f_k^1(X, p + v)$, $S \text{ LexiMax}_v T$ or $S \text{ LexiMin}_v T$. Let us consider four possibilities:

- (a) $f_k^1(X, p) = f_k^1(X, p + v)$. In this case, there exists a $S \in f_k^1(X, p)$ and a $T \in f_k^1(X, p + v)$ such that $S = T$, and makes false $S \text{ LexiMax}_v T$ and $S \text{ LexiMin}_v T$, because the order relations LexiMax_v and LexiMin_v are asymmetric.
- (b) $f_k^1(X, p) = \{C\}$ and $f_k^1(X, p + v) = X_k$. In this case, since $C \in f_k^1(X, p + v)$, there exists a $S \in f_k^1(X, p)$ and a $T \in f_k^1(X, p + v)$ such that $S = T$, and that, as in (a), makes false $S \text{ LexiMax}_v T$ and $S \text{ LexiMin}_v T$.
- (c) $f_k^1(X, p) = X_k$ and $f_k^1(X, p + v) = \{C\}$. It is a case analogous to (b).
- (d) $f_k^1(X, p) = \{C\}$ and $f_k^1(X, p + v) = \{C'\}$ where $C \neq C'$. In this case, when v votes, C loses the Condorcet condition while C' becomes the new Condorcet committee. Since $C \neq C'$ and both C and C' have k elements, $C - C' \neq \emptyset$, $C' - C \neq \emptyset$ and both $C - C'$ and $C' - C$ have the same number of elements. Let us now prove the following claim: “ $y P_v x$ for every $x \in C - C'$ and $y \in C' - C$ ”. Actually, since $x \in C$ and $y \notin C$, x beats y in (X, p) , and, since $y \in C'$ and $x \notin C'$, y beats x in $(X, p + v)$, and it is necessary that $y P_v x$ to change y from being beaten by x in (X, p) to beat x in $(X, p + v)$. We are ready now to prove that $C' \text{ LexiMax}_v C$ and $C' \text{ LexiMin}_v C$. Let C be $\{c_1, c_2, \dots, c_k\}$ where $c_1 P_v c_2 P_v \dots P_v c_k$ and let C' be $\{c'_1, c'_2, \dots, c'_k\}$ where $c'_1 P_v c'_2 P_v \dots P_v c'_k$. Let us consider the number s such that $0 \leq s < k$, $c_1 = c'_1, c_2 = c'_2, \dots, c_s = c'_s$ and $c_{s+1} \neq c'_{s+1}$. If $c_{s+1} P_v c'_{s+1}$, then $c_{s+1} \notin C'$, implying that there exist a $c'_t \in C' - C$ such that $c_{s+1} P_v c'_t$, so contradicting the above claim. Therefore, $c'_{s+1} P_v c_{s+1}$ and this implies $C' \text{ LexiMax}_v C$. In a similar way, let us now consider the number s such that $0 < s \leq k$, $c_k = c'_k, c_{k-1} = c'_{k-1}, \dots, c_{k-s} = c'_{k-s}$ and $c_{k-s+1} \neq c'_{k-s+1}$. If $c_{k-s+1} P_v c'_{k-s+1}$, then $c'_{k-s+1} \notin C$, implying that there exists a $c_t \in C - C'$ such that $c_t P_v c'_{k-s+1}$, so contradicting the above claim. Therefore, $c'_{k-s+1} P_v c_{k-s+1}$ and this implies $C' \text{ LexiMin}_v C$. We can conclude that $S \text{ LexiMax}_v T$ and $S \text{ LexiMin}_v T$ are false. $\square$

Since, as said above, participation in the sense of E implies participation in the sense of every order E' stronger than (or included in) E , Proposition 3 remains valid

if the *LexiMax* or *LexiMin* orders are replaced by stronger extension orders. Thus, the Condorcet k -correspondence f_k^1 satisfies weak *OP*-participation, weak *PE*-participation and weak *LMM*-participation. Thus, f_k^1 is immune to the abstention paradoxes excluded by the weak *LexiMax*, weak *LexiMin*, weak *LMM*, weak *OP* and weak *PE* participation properties.

4 Practical Relevance of the Abstention Paradox

Is the abstention paradox a mere mathematical curiosity? Is it so rare and infrequent that it does not raise any practical concern neither to any voter in the moment of voting, nor to the authority deciding what a voting rule to establish?

One way to help answer these questions is to try to answer what is the probability of occurrence of this paradox.

For the case of Condorcet methods, neither the task of computing the probabilities of occurrence of the paradox in k -choice contexts, nor the task of estimating by simulation the correspondent frequencies in these contexts, have yet, to our knowledge, started. However, there is some research on the estimation and computation of the probabilities of occurrence of the paradox for the conventional 1-choice case. See, for example, [Jimeno \(2003\)](#).

The frequency of occurrence is dependent on the voting method and on the number of voters and candidates.

To be specific, for voting situations of a small-medium size with 5–8 candidates and an odd number of 5–15 voters, [Jimeno \(2003\)](#) shows that some Condorcet methods have a frequency of occurrence of the Optimistic Abstention Paradox (frequency of situations in which the Optimist Participation Property fails) of 8–40%. In such situations there is at least one voter for whom voting has been harmful. In particular, the frequency for the Copeland method is of 8–21%. This is a relatively high frequency, whose practical consequences can not be overlooked.

One could think that the relevant indicator in order to appreciate the practical importance of this paradox is the probability, for any randomly chosen voter, of incurring in any of the different versions of the paradox.²

We agree with this opinion. We can not make a precise estimate of these individual probabilities, because we have not yet done any simulation in order to do such computation. However, we conjecture, based on the global results obtained in previous simulations, that this probability can reach 2% for the Copeland voting rule and even be higher for other Condorcet voting rules.

Since this is still a relatively high probability, specially when compared with the probability for a randomly chosen voter of being pivotal (her vote changes the elected candidate), we conclude that the abstention paradox is not a mere mathematical curiosity, and that the probability of harming oneself when voting in a small-medium sized situation with a Condorcet voting method is small but not negligible.

² We are indebted to an anonymous referee for its suggestions on this discussion, in particular for pointing out the relevancy of the computation of this probability.

It seems intuitively that, as k increases, this probability would decrease. Nevertheless, more work is needed in order to estimate this probability with more precision and for more voting methods in k -choice contexts (even for the most simple case $k = 1$).

5 Concluding Remarks

In this paper we have analysed the Abstention Paradox in Condorcet voting k -functions and k -correspondences.

We have obtained two impossibility results (Propositions 1 and 2) establishing that, for every $k \in \{1, 2, 3, \dots\}$, all Condorcet k -functions fail to satisfy *OP*-participation and (provided they are weak translation invariant) *PE*-participation, and that all Condorcet k -correspondences fail to satisfy moderate *OP*-participation and (provided they are weak translation invariant) moderate *PE*-participation. These results are natural extensions to the k -choice context of the impossibility results in Moulin (1988) for voting functions, and in Jimeno et al. (2009) for voting correspondences.

On the other hand, Proposition 3 is a possibility result establishing that, for every $k \in \{1, 2, 3, \dots\}$, there is a Condorcet k -correspondence satisfying weak *LexiMax* and weak *LexiMin* participation (and consequently satisfying weak *OP* participation, weak *PE* participation and weak *LMM* participation). This proposition establishes the limits to impossibility results and shows that Proposition 1 cannot be strengthened by replacing moderate by weak participation properties.

Another worth considering question is the following: Is it the case that the natural extensions to the k -choice context of known Condorcet correspondences only fall into the mild abstention paradoxes corresponding to the failure to satisfy the participation properties appearing in Proposition 1? Or, on the contrary, fall these k -correspondences into stronger abstention paradoxes? Although the complete answer to this question needs a case by case study, the example above seems to point out to the second possibility. Actually, as the case of MinMax and Copeland illustrate, we could conjecture that failing weak k -*OP*-participation, weak k -*PE*-participation and even weak k -*SD*-participation is a fairly common characteristic of Condorcet k -correspondences.

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